

Chapter 3

Success Stories on the Impact of New Learning Environments on Refugee Students' Writing Skills



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Abstract This chapter explores the transformative impact of innovative educational approaches on refugee students' writing skills, emphasizing the role of AI and new learning environments. It begins by defining writing as a critical skill linked to cognitive and academic success, highlighting its challenges for displaced students. The chapter reviews traditional and process-oriented methods, noting the limitations of rote practices in fostering creativity and meaning. It examines tailored educational models, such as bilingual and metacognitive strategies, and other global examples supporting refugee learners. The discussion extends to AI-assisted tools, particularly ChatGPT®, showcasing their potential to enhance engagement, reduce anxiety, and provide personalized feedback. Practical activities and success stories demonstrate AI's capacity to bridge linguistic gaps and foster inclusivity. The chapter concludes with reflections on integrating these tools into curricula, addressing opportunities and challenges, and advocating for culturally sensitive, technology-driven solutions to empower refugee students.

3.1 Introduction

When discussing the significance of language skills, I often share Kitzhaber's brief anecdote: *"When a primary school student asked what could most effectively improve their language abilities, the teacher's clear response contained an element of truth: 'Sunlight and vitamins'... 'Sunlight and vitamins.'"* This response highlights that the development of language skills is intricately connected to a person's lifelong development and cannot be accelerated independently of that growth

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(Kitzhaber 1963, p. 26). This connection stems from the complexity of language learning and the relationship between fundamental language skills, such as literacy, and individuals' academic, social, and cognitive success (Spinillo and Sotomayor 2023).

Writing, one of the four core language skills, establishes a societal connection between the individual and the written language. For students continuing their education, it makes the core ideas in a text visible and enables them to record, apply, analyze, internalize, and organize those ideas in a lasting way. The fact that many students do not attain adequate proficiency in writing by the time they complete their compulsory education is a significant concern (Ng and Renshaw 2023). This concern becomes particularly pronounced among students born or forced to migrate to a place different from the country where their families were born and raised, along with their cultural values. For these students, it is crucial to develop alternative approaches to encourage writing.

The first part of this chapter focuses on defining writing skills and their role in learning, examining how writing skills contribute to students' cognitive development and linguistic competence. Following this, writing-focused educational programs and models for refugee primary school students will be analyzed alongside their position and implementation processes within the education system. The third section addresses the assessment and evaluation of writing skills, discussing methods and tools used to measure refugee students' writing performance objectively. In the section on writing anxiety and strategies to overcome it, the factors contributing to students' anxiety will be discussed, and practical methods to mitigate this anxiety will be shared.

The chapter will then delve into machine translation and AI-supported methods, which are reinforced with a practical lesson plan. In the final section, AI-assisted writing instruction is explored; specifically, the development of suitable teaching materials and activities for students through AI-supported applications is discussed. The chapter will conclude with examples of AI-based writing practices, accompanied by students' brief reflections and perspectives on these implementations.

This study aims to present current data and success stories regarding the impact of innovative, AI-supported instructional practices designed to enhance refugee students' writing skills compared with traditional methods. By doing so, it seeks to address a significant gap in academic research and practice concerning the education of refugee students while shedding light on the opportunities and challenges that may arise in implementing new learning environments, particularly in writing skills.

3.2 Defining Writing Skills and Their Role in Learning

Writing has been examined by numerous researchers from different perspectives. According to Skinner (1957, p. 69), writing is a verbal behavior akin to speaking. While a speaker creates a visual pattern in the listener's mind, in writing, the visual

stimuli are texts. Moreover, there are various forms of writing, and mastering them requires individuals to develop specific motor skills and learn how to integrate these skills effectively. Processes and social skills must also support the teaching of writing. Teachers play a pivotal role in fostering these social skills and processes (Vygotsky 1986). From Vygotsky's (1986) sociocultural perspective, children internalize the assistance provided by adults or more capable individuals and use it to guide their problem-solving behaviors. Thus, it is essential to support the acquisition of writing skills through social learning.

In Western literate cultures, writing is often regarded as superior to speaking, and writing skills are valued more highly than oral skills. For instance, speaking activities (such as brainstorming and planning) are commonly used in American classrooms as preparation for writing (Nykiel-Herbert 2010). Writing is also a fundamental component of communication and multidimensional thinking. A student's writing ability is a critical factor influencing their persistence and success in any academic context (Stephen 2022). Writing skills, taught and practiced from primary school through the end of one's professional career, are regarded as vital to effective communication (Finley 2021). Individuals with underdeveloped writing skills face disadvantages in securing employment (Harris et al. 2019). Writing is a prerequisite for language teaching and learning and an indispensable language skill for achieving a successful academic career (Razkane et al. 2023). Writing strongly correlates with reading and reading comprehension (Stephen 2024). Students who can comprehend what they read and translate into written language are better equipped to express their thoughts, arguments, or ideas. Meaningful writing involves selecting and organizing ideas within the mind, a process governed by cognitive and metacognitive stages (Knospe 2018).

Traditional writing strategies have typically focused on the final product, i.e., the written text itself (Chien 2012; Khosravi et al. 2023), emphasizing its mechanical aspects such as punctuation, spelling, grammar, and syntax (Erol and Kavruk 2021). Numerous methods serve this purpose. One such method involves copying sentences or texts verbatim (de Rose 2023). This approach is still widely used in Türkiye under the new curriculum implemented in 2024 to teach students how to write. While copying a text can effectively enhance motor skills related to writing, it is often insufficient on its own for developing more advanced writing skills. Copying a text may fall short of fostering meaningful writing abilities. The minimum skills required for copying a text are formally like those needed for dictation (de Rose 2023). Although dictation reinforces the accurate writing of letters, it does not contribute to the development of semantic understanding.

Another writing strategy is the phonological method, which associates sounds with letters. In this approach, the teacher verbally expresses a text or sentence, and the student transcribes the words they hear. This method is particularly effective in helping students translate spoken words into written form (Civelekoğlu et al. 2024). By practicing letter-sound matching, individuals can increase their phonological awareness while improving their word-reading and transcription skills. However, this method often fails to foster meaningful writing or the ability to make sense of written content such as dictation. Traditional strategies overlook the concepts,

events, emotions, and thoughts in an individual's cognitive universe, which are transformed into text through various cognitive processes (Erol and Kavruk 2021).

While traditional methods underline the foundation of basic writing skills, they seem to lack addressing the cognitive and creative aspects of writing. This has led to the emergence of process- and cognition-oriented approaches. In recent years, writing strategies have evolved toward a more process- and cognition-oriented approach (Khosravi et al. 2023; Razkane et al. 2023; Yang et al. 2024). Metacognition permeates the writing process, during which students plan, monitor, and evaluate their cognitive, behavioral, and emotional skills (Xu and Zhu 2024). Metacognitive strategies are critically important in writing (Feng et al. 2021; Haukås 2018; Knospe 2018; LV and Chen 2010; Zhang and Qin 2018) because proficient writing skills depend not only on linguistic processing but also on metacognitive knowledge and regulation (McCormick 2003). Metacognitive writing strategies offer a multidimensional writing process as they involve reflecting on the learning process, planning to learn activities, monitoring comprehension or production during the process, and conducting self-assessment after completing the task (Chamot and O'Malley 1987, p. 20). Since metacognitive strategies may help bridge gaps in linguistic and cultural knowledge, they can be transformative for refugee students, as well.

In writing, transforming ideas into written language requires the writer to generate new ideas while maintaining text coherence to ensure that the writing is logical and cohesive. Writers who can review and monitor their work tend to exhibit more advanced metacognitive skills than their peers do (Soto et al. 2022). However, teaching metacognitive strategies demands significant time investment (Khosravi et al. 2023). Although teachers emphasize metacognitive strategies in traditional classroom settings in detail, students often fail to complete the writing process during class. As a result, students are required to continue and complete the process independently outside of class. These out-of-class processes occasionally present challenges in the consistent application of these strategies.

3.3 Writing-Focused Educational Programs and Models for Refugee Primary School Students

The ability to recognize and write words in a language is expected to develop gradually after the letters of the alphabet and basic vocabulary are mastered. All beginner-level reading and writing models transition between reading and writing simple, short texts through several stages (Seymour et al. 2003). The first stage involves recognizing and memorizing prominent cues in the text, allowing individuals to identify words. They subsequently learn to write the letters corresponding to the sounds that those words represent, thereby mastering basic decoding. In the next stage, this foundational decoding skill extends to more complex words, culminating in the final stage of reinforcement and fluency. Importantly, beginning to write reinforces learning to read (Boon 2014; Nassaji 2007).

For second-language learners, while it is often observed that individuals who are literate in their native language often experience fewer difficulties when learning a new language, particularly at the primary school level (UNHCR [United Nations High Commissioner for Refugees], 2024), challenges may arise with some words in languages where letters or letter clusters have multiple pronunciations and spellings. This is especially true during dictation and spoken language practice in English, French, or Danish. However, this issue is not universal; in some languages, words are almost always written as pronounced (e.g., Italian, Turkish). Differences in linguistic characteristics can also lead to variations in the time required to learn written language skills.

A study concluded that the orthographic characteristics of languages significantly influence writing skills. The research revealed that in languages where most words are written as they are pronounced—such as Greek, Turkish, Finnish, German, Italian, and Spanish—children could easily write what they read by the middle of the first grade. Conversely, children demonstrated significantly lower writing performance in languages such as English, where there is a lack of correspondence between spelling and pronunciation, achieving only 34% accuracy. Similarly, children learning to write in other non-phonetic languages presented low writing proficiency, with accuracy rates of 71% in Danish, 73% in Portuguese, and 79% in French (Goswami et al. 1998). Comparable findings have also been reported in other studies (Frith et al. 1998).

These differences in the orthographic features of native languages may contribute to variability in the writing development of refugee children learning a second language (Ziegler and Goswami 2006). However, despite these differences in reading and writing development across languages, the “interdependence hypothesis” posits that the time refugee students dedicate to developing writing skills in their native language does not negatively impact the time they spend learning the host country’s majority language. In other words, the hypothesis suggests that proficiency in a refugee student’s native language does not adversely affect their learning of the majority language in their host country (Genesee et al. 2006).

Developmentally, access to language education for refugee families yields better cultural integration, academic achievement, and employability outcomes for all family members. However, without implementing supplementary literacy and integration programs for school-aged children, refugee students often face challenges, particularly in reading and writing (UNHCR 2024). Some countries have established specialized writing and reading courses to address these issues. For example, the European Union-funded programme *Promoting Inclusive Education for Kids in the Turkish Education System (PIKTES)* has been implemented in Türkiye. Through PIKTES, refugee students are expected to acquire the following writing skills (PIKTES 2024):

- Newly learned words are written, and phrases are set.
- To transcribe words and simple sentences they hear.
- To compose sentences containing basic question structures.
- To create simple sentences because of visual cues.
- To describe a visualized event in their own words.

The goal is to develop these skills in primary school refugee students by practicing short writing exercises. The suggested topics for these exercises include the following:

- Simple texts involving directions and address descriptions.
- Texts containing frequently used vocabulary for daily life.
- Texts focused on personal care and health expressions.
- Texts related to hobbies, interests, or fears.
- Texts involving vocabulary for shopping and personal needs.
- Descriptive texts based on provided visuals.

These targeted activities aim to support refugee students in gradually developing essential writing skills within a structured framework tailored to their unique needs.

Some studies (Goswami et al. 1998; Frith et al. 1998) have highlighted the challenges primary school students face in learning English as a second language, particularly for those considering English acquisition challenging/as a challenge. To support English instruction, the volunteer organization HIAS provided free supplemental lessons to refugee primary school students at a public elementary school in Pittsburgh (HIAS 2024). Refugee children from Afghanistan, the Democratic Republic of Congo, Syria, and Somalia who had been in the United States for 5 months to 2 years participated in this program. Over 5 months, they received literacy training. This program's reading and writing materials focused on words commonly used in daily life and content aimed at facilitating adaptation to American culture. Additionally, texts addressing the prevention of peer bullying were included. Many volunteer organizations offer free supplemental language lessons in the United States, particularly on developing reading and writing skills.

Different countries implement various reading and writing models or additional programs for refugee students, and small-scale academic studies on program writing can be found. For example, Nykiel-Herbert's (2010) study with Iraqi refugee primary school students in the United States involved an intervention program designed to improve their literacy skills. This program provided full-day instruction in a separate classroom tailored to Iraqi refugee students' experiential knowledge, cultural norms, values, beliefs, and expectations. While the reading and writing activities were conducted in English, the students were encouraged to use their native languages (Kurdish and Arabic) to collaborate with classmates and facilitate easier communication. According to the results of the reading/writing test administered at the end of the program, Iraqi refugee primary school students showed significant progress compared with other English learners. When their reading and writing skills were compared internally, their reading skills improved more significantly than their writing skills did.

Cranitch (2010) conducted a similar literacy program with Sudanese refugee students in Australia. The *Literacy Transition Pilot Programme* (LTTP) was implemented at an intensive English center with a curriculum designed to address students' background knowledge gaps. The program was conducted with a small group of students in a full-day format within a classroom that included a Sudanese

assistant teacher to facilitate communication and help students express themselves more comfortably. The program's primary goal was to enhance reading and writing skills through specialized instruction. The findings demonstrated significant progress in the language development of the participating students, as measured by English as a second language (ESL) proficiency scale. Another noteworthy outcome of this study, consistent with Nykiel-Herbert's (2010) findings, was that students showed more significant improvement in reading skills than writing skills.

3.4 Assessment and Evaluation of Writing Skills

The development and accurate evaluation of writing skills are essential to refugee students' academic success, written communication competence, and ability to express themselves in writing. In recent years, assessing and evaluating refugee students' writing skills has become a significant area of research for educators, researchers, and policymakers. Writing skills are critical in understanding refugee students' language proficiency and supporting their academic progress (Hughes 2003).

When designing or selecting assessment tools for writing skills, it is important to consider that the target group consists of refugee students. These students may have traumatic backgrounds, face cultural differences, lack prior educational experience, and be in the process of learning a language other than their native one (Reed et al. 2012). Furthermore, adapting to a language different from their native language and producing texts in an academic writing format can pose significant challenges for refugee students. Traditional assessment methods often fail to capture these students' individual achievements and struggles (Harklau 2002). The conventional methods for assessing writing skills include the following:

- Structured multiple-choice tests,
- Essay-writing exams on specific topics,
- Tests directly measuring grammar and vocabulary knowledge,
- Assessments focus on spelling and punctuation exercises (Siddiq et al. 2016; Weigle 2002).

The writing skills of refugee students should be assessed using alternative methods that consider their unique challenges and circumstances. These non-traditional approaches focus on evaluating the use of writing in real-life contexts. Some alternative methods for assessing writing skills include the following:

- Creating and evaluating portfolio files containing products developed throughout the process,
- Writing journals on various topics,
- Peer evaluations and self-assessment forms,
- Project-based writing tasks,
- Writing activities include letter writing, email composition, and storytelling (Green 2022; Matsuda and Snyder 2017; Warschauer 2006).

In addition to these alternative methods, technology-supported assessment and evaluation approaches are increasingly being researched and implemented. Technology-based writing assessments offer more flexible and culturally appropriate environments for refugee students (Engin et al. 2024; Zuo and Ives 2024). For example, AI-supported assessment activities, automated writing evaluation systems, and some paid writing applications can effectively measure refugee students' individual progress (Hooda et al. 2022; Kayali and Balat 2024). Also, AI-supported systems can provide personalized recommendations and corrections to promote students' proficiency levels. For refugee students, this offers an opportunity for self-paced learning in a non-judgmental environment, and this may reduce anxiety associated with traditional assessments. Whether traditional, alternative, or technology-based assessment methods are used, the cultural sensitivity and appropriateness of the tools employed in evaluating refugee students' writing skills are critical. Specifically, the types of texts, topics, and instructions provided should align with the student's cultural contexts. Additionally, educators conducting the assessments must be free from cultural biases to ensure fair and accurate evaluations.

3.5 Writing Anxiety Among Refugee Primary School Students and Strategies for Overcoming It

Writing anxiety could also be discussed as a struggle to manage language demands and the emotional fear of failure, which may hinder both fluency and creativity in writing (Bloom 1985). Responses such as sweating, avoidance of writing, and fear of negative evaluation are commonly associated with writing anxiety (Sabti et al. 2019; Zorbaz 2011). When students experience writing anxiety, they often focus on minor details rather than the main idea of the text, which negatively impacts their writing performance (Rasool et al. 2023). Furthermore, since writing skills are still developing in early primary school students, their levels of anxiety may be higher, even in the absence of specific negative experiences (Hidi and Boscolo 2006).

Writing anxiety is particularly important for refugee primary school students learning a language other than their native language, as it can have a direct effect on their entire educational journey (Dornyei 2005; Qashoa 2014). The trauma, language barriers, and challenges of cultural adaptation these students face contribute to more frequent and intense writing anxiety. When writing in a new and second language, they must contend with emotional and cognitive difficulties (Suárez-Orozco et al. 2008). Krashen's *affective filter hypothesis* underscores that reading and writing anxiety, motivation, and other affective factors significantly influence second-language acquisition. The hypothesis asserts that a low-stress learning environment is essential for fostering progress in a second language (Krashen 1982).

There is a limited understanding of the relationships among bilingualism, writing anxiety, and cognitive load (Lee et al. 2022). In this context, cognitive load refers to the mental burden or strain on an individual's cognitive system while performing a learning task (Sweller et al. 1998). Writing anxiety can be attributed to language barriers, emotional factors, and environmental influences. Language barriers often involve deficiencies in spelling, grammar, and vocabulary (Krashen 1982), whereas emotional factors include fear of criticism and negative reactions from peers (McBrien 2005). Environmental influences may include inadequate learning environments and teachers' failure to address refugee students' language needs (Pinson and Arnot 2007).

The Council of Europe's guide on second-language learning for immigrants outlines strategies to prevent writing anxiety. Refugee students may benefit from affective or socio-interactive strategies to overcome speaking or writing anxiety (Council of Europe 2022).

Affective strategies:

- Encouraging positive self-talk, fostering motivation, practicing deep breathing, and cultivating positive thoughts to enhance writing or speaking motivation.
- Using tools to reduce anxiety, such as calming music, artistic works, or humorous videos.

Socio-interactive strategies:

- Practicing writing or speaking with someone they feel comfortable with (e.g., a peer, teacher, or mediator).
- Anxiety-reducing and interactive tools such as artificial intelligence, translation machines, or online dictionaries are employed.

Additional strategies for reducing writing anxiety include psychosocial support strategies and other approaches to promote writing development.

Psychosocial support strategies:

- Teachers adopt trauma-sensitive educational techniques, such as giving aggressive or shy students time to adjust, rewarding positive social behaviors, and actively listening to students (Cole et al. 2005).
- Providing supportive and motivating feedback on students' writing products instead of offering criticism (Graham and Perin 2007).

Strategies to promote writing development:

- Implementing programs that support bilingualism (Cummins 2001).
- Creating a comfortable, supportive, and creative writing environment in the classroom (Fletcher and Portalupi 2001).

The writing process is modeled by providing students with sample texts (Graham and Perin 2007).

3.6 Internet-Based Methods for Increasing Refugee Students' Writing Skills

Writing is not an innate skill (Kartawijaya 2018); instead, it is a skill that requires practice, development, and reflection. To improve writing skills, frequent practice is essential. With enhanced writing ability, individuals can produce effective and well-structured texts (Amyatun and Kholis 2023). Academic writing within the educational context is distinct from and more challenging than everyday writing. Teachers must employ appropriate writing techniques to prevent students from struggling with writing. Traditional teaching approaches, however, can lead to boredom, especially for refugee students who face language barriers and may hinder their writing proficiency. As refugee students face significant barriers in developing writing skills due to linguistic and cultural differences, integrating technology such as machine translation (MT) tools provides an opportunity to bridge these gaps.

Recent studies have increasingly focused on computer-assisted language instruction in today's digitized classrooms, moving away from traditional teaching methods (French 2016; Giannetti 2016; Krulatz and Iversen 2019; Tsai 2019). Writing tasks supported by MT tools (e.g., Google Translate®, DeepL®) or artificial intelligence systems (e.g., ChatGPT®, Gemini®) are generally studied under the framework of *computer-assisted language Learning* (CALL) (Fredholm 2019). Research on CALL in second-language contexts has found that computer-assisted instructional methods improve students' writing quality and that participants exhibit positive attitudes toward the use of these tools as educational aids (Fredholm 2019; Kelly and Hou 2022; Lee 2020; Tsai 2019; Su et al. 2023). However, it has also been noted that MT applications may have pedagogically adverse effects, mainly due to grammatical errors in translations between the native and target languages (Arnot et al. 2014; Machida 2011; Tasdemir et al. 2023).

This highlights the need for careful and informed technology integration in teaching writing skills to ensure its benefits while mitigating potential drawbacks. When discussing MT applications, the first and most widely recognized tool that comes to mind is *Google Translate® (GT)*, a free machine translation service. GT facilitates the translation of documents and texts between languages and is pre-installed on many modern mobile devices, making it readily available to users. Given its widespread availability, teachers and students often turn to GT when they need to learn the spelling or meaning of a word in a particular language. Although GT is primarily a web-based tool, it can also be used offline when language files are downloaded in advance, which makes it particularly convenient for use in school settings.

In educational environments, GT is frequently employed to assist students in learning one or more languages while completing their writing assignments. Additionally, it is used for translating foreign language texts, understanding study materials in other languages, and even supporting the essay-writing process (Perdana and Arifani 2024). Notably, using.

GT can indirectly foster students' curiosity. Refugee students, in particular, who are constantly exposed to new words in the language of their host country, may find themselves drawn into a linguistic exploration, engaging actively with the target language. A literature review of GT reveals that few studies have focused on primary school students. One notable study using GT as a learning tool to enhance native language writing skills in primary school was conducted by Stapleton and Kin (2019). In their research involving 26 eleven-year-old primary school students in Hong Kong, the participants were asked to complete an argumentative writing task via GT. The task was assessed based on grammar, vocabulary, and comprehensibility criteria. The findings showed that teachers gave significantly higher scores for grammar in texts written with GT than in those written without it. However, no significant difference was observed in vocabulary. Regarding comprehensibility, texts produced with GT were more coherent and understandable. Despite these insights, no studies have explored the use of GT or other MT tools to teach or improve language skills in refugee primary school students.

Although MT has a long history, its use in second-language writing instruction was rare until recently. This limited adoption is primarily attributed primarily to inaccuracies in earlier MT translations (Briggs 2018; Lee 2023), which were characterized by errors in grammar, spelling, and syntax (Garcia and Pena 2011). Recognizing these shortcomings, GT introduced a significant update in 2016, implementing Google's Neural Machine Translation (GNMT), a neural network approach for analyzing word sequences. This update claimed that the GNMT achieved accuracy levels in some tests comparable to those of bilingual human translators (Wu et al. 2016).

However, over-reliance on MT tools or a reluctance to write without them may hinder the development of students' original writing skills and could, at times, lead to academic integrity violations (Fitriani et al. 2021). Additionally, the frequent use of MT tools in classroom activities may foster dependence on mobile devices or computers. To prevent over-reliance, educators can design activities that require students to draft initial versions without technology, followed by using MT tools for editing and refinement. Further research into AI-supported translation programs is essential to maximize the efficiency of MT tools. Within this framework, a lesson plan utilizing MT to enhance the writing skills of refugee primary school students is presented below:

– *Writing Activities Lesson Plan 1*

Unit: How and Why Should We Use Machine Translation to Teach Writing Skills to Refugee Primary School Students?

Level: A2

Objectives:

1. The main ways in which machine translation (MT) can be used in the context of written communication are introduced.
2. The findings can help students understand the impact of machine translation on the development of writing skills.

Learning Outcomes:

1. Can compose an introduction to a story or continue a story, provided they can consult a dictionary and references (e.g. tables of verb tenses in a course book).

Methods and Techniques: Presentation and discussion.

Lesson Stages.**Part 1**

Preparation (10 min):

1. Purpose and Context:

- Define the role of machine translation (MT) technology in improving children's writing skills, focusing specifically on the challenges refugee students face in overcoming language barriers.

2. Introduction to Machine Translation:

- A brief demonstration of tools such as Google Translate, DeepL, or similar applications.
- The role of MT in the writing process with input from the class is discussed.

3. Discussion Questions:

- "Have you ever used machine translation while writing? If yes, what was your experience like? Can you express your experience visually, in writing, or verbally?"
- Brief student responses are collected.

Part 2

The Role of Machine Translation in the Writing Process (25 minutes)

1. Presentation: Advantages and Limitations of Machine Translation

- **Reducing language barriers:** Highlight its role as a bridge between native and target languages.
- **Expanding vocabulary:** Explaining how vocabulary enables students to learn new words and expressions.
- **Supporting written expression:** Emphasize its role in showcasing proper sentence structures.
- **Grammar errors:** Use visuals to show that MT may not fully reflect the nuances of the target language and could introduce inaccuracies.

2. Activity: Hands-On Writing Exercise

- Depending on the availability of computers in the classroom, the teacher can project their computer screen or allow students to use individual devices to access an MT application.
- Example sentence: "Today, we played a very fun game at school."
- Students will translate this sentence from their native language to the target language (e.g., from Arabic to Turkish and then back to Arabic).

- Students review the output and evaluate its accuracy together as a class.
- After the evaluation, the class will generate a new sentence related to the original one, translating it as well.
- Together, students create a short text/story based on the sentences, translate it via MT, revise it as needed, and finalize the text/story after collaborative review.

Part 3

Evaluation (10 minutes)

1. Discussion Questions

- “Would you consider using MT in the future? Why or why not?”
- “What could be done to use MT more effectively?”
- “Did you enjoy the writing activity using MT?”

This lesson plan offers a structured approach to integrating machine translation into the classroom. It helps refugee primary school students develop their writing skills while fostering an understanding of the benefits and limitations of MT tools.

3.7 AI-Based Writing Skills Instruction

Artificial intelligence (AI), which offers users unparalleled data and problem-solving capabilities, has become widely used in numerous domains such as manufacturing, economics, transportation, healthcare, communication, and education. With the growing societal impact of AI, educators have recognized the urgent need to incorporate AI into K-12 education by developing programs and conducting research to encourage its use among students (Cantú-Ortiz et al. 2020; Zhang et al. 2024). A joint report by UNICEF and the World Economic Forum highlights how AI influences children’s rights and everyday experiences, noting that as AI technology advances, it presents immense opportunities and challenges. While AI-powered intelligent systems can potentially enhance children’s learning experiences and educational outcomes, concerns such as data collection and analysis, identity theft, harmful content generation, and safeguarding children’s privacy and security must not be overlooked (UNICEF 2023a, b).

In recent years, teachers’ attitudes toward and understanding of AI have directly impacted their use of AI tools in the classroom (Li and Ironsi 2024). Professional development programs, lesson plans, and example applications should be provided to enable teachers to effectively integrate AI into their lessons while maintaining a balance and ensuring children’s safety (Elbanna and Armstrong 2023). This is particularly critical given many teachers’ apprehension upon encountering ChatGPT—one of the most prominent examples of AI—and realizing its capabilities. A key concern was that students might use ChatGPT as a shortcut to avoid the challenging process of writing practice, treating it as a “cheating robot” (Cotton et al. 2024; Levine et al. 2024).

ChatGPT, developed by OpenAI, offers new opportunities and challenges for students and teachers in their approach to various educational materials. Its advanced language models can generate human-like responses to text-based inputs. With the anticipated growth of AI in education, it is crucial to investigate the role of ChatGPT in the learning process and to support teachers in integrating this new technology into educational environments (AlAfnan et al. 2023; Levine et al. 2024). From a student perspective, existing studies acknowledge AI's contributions to enhancing learning, providing personalized education, and improving the overall learning experience (Fauzi et al. 2023; Li and Ironsi 2024). Since 2022, schools, teachers, and researchers have lifted bans on using ChatGPT and begun experimenting with its integration into writing instruction (Banks 2023). ChatGPT has been proposed as a writing coach or a tool to inspire students during the writing process. It can support writing skills by reducing the cognitive effort, space, and time required for composing texts (Daniel et al. 2023; Kleiman and Gallagher 2024). With AI's growing role in education, particularly in writing instruction, it is essential to navigate its potential while addressing its challenges, ensuring it serves as a tool to enhance teaching and learning outcomes.

3.8 Success Stories in Developing Writing Skills with AI

While various versions of AI are currently used in education, this section highlights the success of employing ChatGPT as an AI-assisted writing tool with refugee primary school students (Applicable at A1 / A2). The first part provides a brief overview of the students' demographics and the school where the tool was implemented. The second part describes how AI-assisted writing applications were conducted, and the final section shares students' feedback on these applications and their impact on their writing skills. The purpose of this example is to illustrate how refugee students can enhance their writing skills with the assistance of artificial intelligence. Although the example was conducted in Türkiye, it was designed to apply to refugee students globally. Within the framework of the Common European.

Framework of Reference for Languages (2020) aims to develop creative writing skills, encompassing the following learning outcomes:

- Can produce simple phrases and sentences about themselves and imaginary characters, where they live, and what they do.
- Can use simple words/signs and phrases to describe certain everyday objects (e.g. the colour of a car, whether it is big or small).
- Can compose an introduction to a story or continue a story, provided they can consult a dictionary and references (e.g., tables of verb tenses in a coursebook).
- Can give very short, basic descriptions of events, past activities and personal experiences.
- Can tell a simple story (e.g. about events on a holiday or about life in the distant future).

3.8.1 Part I: General Overview of the School

Over the past 6 years, Türkiye has become the country hosting the most significant number of refugees worldwide, with over 1.7 million refugee children (UNICEF 2023a, b). A significant portion of this population is concentrated in the cities bordering Syria and Iraq, such as Gaziantep, Hatay, and Kilis. Refugee students and their teachers face substantial challenges due to linguistic, cultural, and instructional barriers in the target language. Most of these students are children of Syrian or Iraqi families displaced for political reasons. The limited Turkish proficiency of their families hinders students' ability to overcome language barriers at home and in school, affecting their ability to complete assigned activities successfully. According to Turkish education unions, certain misconceptions regarding individuals who speak languages other than Turkish lead to inconsistencies in the activities designed to teach Turkish to native Arabic or Kurdish speakers (Turkish Education Syndicate 2024).

For this project, I worked with a primary school in the southwest of Gaziantep in Türkiye, which serves approximately 1000 students. Over 70% of the student body (860 out of 1200 students) consists of bilingual refugee students whose first language is Arabic or Kurdish (Turkish Education Syndicate 2024). While native language instruction is provided, all other subjects are taught in Turkish. Special emphasis is placed on teaching Turkish as the target language. After regular school hours, classrooms are allocated to refugee students to develop their language skills through targeted activities. Teachers at schools employ various instructional methods to help all students comprehend the educational content while bridging communication gaps. The AI-supported writing study involved fourth-grade students (A1 / A2) aged 9–10 years who were born in Syria (Student 1) and Iraq (Student 2) but displaced shortly after birth and began their education in Türkiye. The selection criterion for participants included having at least basic computer literacy.

3.8.2 Part II: AI-Assisted Writing Activities

The participating school actively supports refugee primary school students in improving their language skills by allowing them to use all available resources outside regular school hours. As part of this initiative, the school's laboratory was accessible every hour after class.

Students were given access to internet-enabled computers in the school's computer lab, and I asked the refugee students to open the ChatGPT application. I chose ChatGPT as a writing assistant because its advanced language capabilities allow researchers, writers, and especially students to improve their writing, edit their work, or correct errors. This made it an ideal tool for helping refugee primary school students develop their writing skills.

We began with simple sentence structures and examples from daily life to help the students into write in Turkish. I opened ChatGPT on the teacher's computer in the lab and entered the following prompt:

Prompt:

Hello. I am a bilingual primary school student. I know some Turkish and some Arabic, but my native language is Arabic. Can you help me improve my Turkish writing skills, which is the language I use where I currently live?

ChatGPT:

Hello! Of course, I'd be happy to help. When starting to write in Turkish, practicing by forming simple sentences and using examples from daily life is very helpful. The following steps can guide you as you write in Turkish.

After this brief exchange, the writing assistant begins generating activities that the students can practice, eventually producing six different activities. The first activity was as follows:

– **Activity 1: Basic Sentence Patterns** ChatGPT

- “My name is ... - اسمي هو ...” (Ör: Benim adım Ali/ My name is Ali)
- “I am ... years old. - ... سنوات” (Ör: Ben 8 yaşımdayım. I am 8 years old)
- “I like ... - ... seviyorum” (Ör: Ben futbol oynamayı severim. I like playing football)
- “I live in ... - ... أنا ... على قيد الحياة” (Ör: Ben İstanbulda yaşıyorum. I live in İstanbul) The students were instructed to fill in the provided sentence patterns with their own information.

I guided them with the following instructions: “Now, open the writing assistant on your own computers and fill in the blanks in the activity with your personal information. After completing the sentence, add the question ‘Did I write this correctly in Turkish?’ at the end of your prompt.”

Student Example Prompt: *Benim adım Ahmed, Türkçe doğru yazdım mı? (My name is Ahmed, did I write this correctly in Turkish?)*

ChatGPT Response: *Yes, the sentence ‘Benim adım Ahmed.’ is correctly written in Turkish. There are no issues with your writing. You could also write it like this:*

- “Adım Ahmed.” (I am Ahmed)
- “Ben Ahmed’im.” (I’m Ahmed)
- Both are correct and understandable expressions in Turkish.

All the participants successfully wrote their sentences correctly. If there had been any spelling or word errors, ChatGPT would have provided feedback and guided them until they corrected the mistakes. This activity took approximately 30–35 min to complete. After a short break, we began the second lesson and proceeded to the second activity generated by the writing assistant.

– **Activity 2: Frequently Used Words**

ChatGPT: *Some frequently used words in Turkish and their meanings are as follows:*

- **Okul** (مدرسة) (School)
- **Öğrenci** (طالب) (Student)
- **Öğretmen** (معلم) (Teacher)
- **Kitap** (كتاب) (Book)
- **Kalem** (قلم) (Pen)
- **Arkadaş** (صديق) (Friend)
- **Oyun** (لعبة) (Game)
- **Yemek** (طعام) (Food) *You can use these words in your new sentences.*

In the second activity, the students were instructed to compose simple sentences using the keywords provided by the writing assistant. They were further directed to add the question “*Did I write this correctly in Turkish?*” at the end of their prompt.

Student Example Prompt: *Okulumu seviyorum, Türkçe doğru yazdım mı? (I love my school; did I write this correctly in Turkish?)*

ChatGPT Response: *Yes, the sentence ‘Okulumu seviyorum.’ is correctly written in Turkish. You’ve constructed a beautiful sentence! If you’d like to add more detail, you could extend your sentence like this:*

- *‘Okulumu seviyorum çünkü arkadaşlarım çok iyi.’ (I like my school because my friends are very good)*
- *‘Okulumu seviyorum çünkü öğretmenim bana yardımcı oluyor.’ (I like school because my teacher helps me)*
- *‘Great job, keep going!’*

At this stage, the positive reinforcement provided by the writing assistant brought smiles to the students’ faces. Some students continued writing; for example, one wrote, “*I love my school and my friends,*” while another wrote, “*I love my school and you.*” They engaged in a dialogue with the writing assistant for a while gradually treating ChatGPT as if communicating with a human. Occasionally, the assistant provided gentle corrections for minor word errors, helping the students refine their sentences. The second activity, which took approximately 20 min, was completed after the students composed sentences using all the provided keywords. Since the students were actively engaged and did not want a break, we proceeded directly to the third activity prepared by the writing assistant.

– Activity 3: Modifying Example Sentences ChatGPT:

1. “*Bugün okulda çok eğlendim.*” (I had a lot of fun at school today.)
(اليوم استمتعت كثيراً في المدرسة)
2. “*Öğretmenim bana yeni bir kitap verdi.*” (My teacher gave me a new book)
(أعطاني معلمي كتاباً جديداً)
3. “*Arkadaşlarımla bahçede oyun oynadım.*” (I played with my friends in the garden)
(لعبت مع أصدقائي في الحديقة)
4. “*En sevdiğim ders Matematik.*” (My favorite subject is mathematics)
(مادتي المفضلة هي الرياضيات)

You can change and write these sentences according to your own life.

In the third activity, the students were asked to modify the given example sentences according to their preferences and rewrite them. For example, they were instructed to replace “*My favorite subject is mathematics*” with their own favorite subject (e.g., “*Prompt: My favorite subject is music. Did I write this correctly in Turkish?*”). The writing assistant provided feedback to the students on the basis of their responses: correct sentences received feedback, as shown in *Example 1*, whereas incorrect sentences were addressed with feedback such as *Example 2*. This activity lasted approximately 20 min. Although the students reported not being tired, the first day of activities was concluded.

Example 1

Student Example Prompt: *En sevdiğim ders müzik, Türkçe doğru yazdım mı?* (My favorite subject is music, have I spelled Turkish correctly?)

ChatGPT Response: *Yes, the sentence ‘En sevdiğim ders müzik.’ (My favorite subject is music) is completely correct! Great job—no errors at all! If you’d like, you can extend your sentence by explaining why you like music:*

- *‘En sevdiğim ders müzik çünkü şarkı söylemeyi çok seviyorum.’ (My favorite subject is music because I love to sing)*
- *‘En sevdiğim ders müzik çünkü yeni melodiler öğreniyorum.’ (My favorite subject is music because I learn new melodies)*

‘Keep it up, you’re doing great!’

Example 2

Student Example Prompt: *En sevdiğim ders mizik, Türkçe doğru yazdım mı?* (My favorite subject is music. Did I spell Turkish correctly? – Misspelled the word “music”)

ChatGPT Response: *Your sentence is lovely, but we can make a few small corrections: Correct version: ‘En sevdiğim ders müzik.’*

- Correction:**
1. *Replace ‘mizik’ with ‘müzik’ (the letter ü should be used).*

After this correction, your sentence is perfect! Keep going; your writing is improving beautifully!”

– Activity 4: Editing a Short Paragraph

ChatGPT: *I’ve written an example paragraph for you below. Try to rewrite it by changing the sentences to fit your own story:*

Hello, my name is Ali. I am 9 years old. I am in the third grade of primary school. I love my school and my friends. My favorite subject is Turkish because I enjoy learning new words. My teacher is very kind and helps me with everything.

You can modify the names, numbers, and content in this paragraph to reflect your own story.

From the first to the fourth activity, the writing assistant demonstrated a teaching method that progressed from parts to the whole. The first activities focused on words, followed by sentences, and by the fourth activity, the focus shifted to paragraph writing. The students were asked to update the paragraph with their own information and input their version into ChatGPT.

Example

Student Example Prompt: *Hello, my name is Mohammed. I am 10 years old. I am in the fourth grade of primary school. I love my school and my friends. My favorite subject is sports because I love playing football. My teacher is very kind and helps me with everything. Did I write this correctly in Turkish?*

ChatGPT Response: *Yes, you've written wonderfully! Your paragraph is entirely correct. There are no spelling errors! Your sentences are clear and beautifully expressed. If you keep this up, your Turkish will continue to improve. Congratulations!*

In this activity, the writing assistant provided constructive feedback by correcting any errors and offering positive reinforcement. Notably, after the fourth activity, the writing assistant no longer included Arabic in its responses, encouraging students to write entirely in Turkish. The paragraph-writing activity lasted approximately 30 min. Some students expressed interest in reading their paragraphs aloud, so a reading session was held with those who wished to participate, lasting an additional 20 min. This concluded the first lesson of the second day. By the end of the fourth activity, I observed a marked increase in students' enthusiasm for writing. Some students even expressed a desire to interact with the writing assistant beyond the given activities. Interestingly, a few students chose to skip their break time to continue practicing with ChatGPT.

– Activity 5: Creating a Paragraph

ChatGPT: *Now you can create your own paragraph. When writing, I recommend paying attention to the following points:*

After a paragraph is written, it is checked via the following steps:

1. *Are the sentences clear? (Is the word order within each sentence correct?)*
2. *Are there any spelling mistakes? (Ensure each word is spelled correctly.)*
3. *Are the punctuation marks correct? (Don't forget to use a period at the end of a sentence and a question mark when asking a question.)*

By following these steps, you can improve your writing.

I anticipated that students would find this activity challenging because, while the writing assistant had previously helped them develop their writing skills by guiding

them to make minor modifications—similar to the dictation method—this task required them to write a paragraph from scratch without prompts. My predictions were accurate; nearly all the participants attempted to replicate the paragraph from the fourth activity or asked the writing assistant to generate a new example for them. This also demonstrated that they were beginning to understand how to use AI effectively.

At this point, I asked the students to close their computers and write their paragraphs in their notebooks. Writing new paragraphs in their notebooks took significantly more time. While they were writing, I provided them with some hints. Once they had finished, I asked them to review their paragraphs following the steps outlined by the writing assistant and then input their paragraphs into ChatGPT for feedback.

Example

Student Example Prompt: *“One day, Ali went to the forest and made friends with a bird. The bird chirped at him, and Ali thanked it. The bird showed Ali its friends and said, ‘These are my friends.’ Ali was very excited. It was a wonderful day. Did I write this correctly in Turkish? Is the story meaningful?”*

ChatGPT Response: *Your sentences are generally correct in Turkish, and the story is meaningful. With minor punctuation or fluency adjustments, you can express your story even better. Here are some suggestions:*

1. **Revised Version:**

One day, Ali went to the forest and made friends with a bird. The bird chirped at him, and Ali thanked the bird. Later, the bird showed Ali its friends and said, ‘These are my friends.’ Ali was very excited. It truly was a wonderful day.

- Replaced ‘Ali thanked it’ with ‘Ali thanked the bird’ to enhance fluency.
- The dialog was placed in quotation marks for clarity.

2. **Additional Suggestions:**

- If you want to write a longer story, you could add details about the bird’s friends, what they did, and how Ali felt.
 - Using quotation marks for dialog makes the story easier to follow:
- *‘Hello, Ali!’ chirped the bird. Ali smiled and asked, ‘Hello, bird, what’s your name?’*

Your text is grammatically correct and meaningful as a story. With small adjustments, you can enrich your story even further. Well done!

By the end of this full-day activity, it was clear that the students had become quite comfortable with the writing assistants. After spending 3 days working

(continued)

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with prompts such as the one above, we continued using new activities generated by the assistant (e.g., journaling, picture description, digital storytelling, word-sentence-paragraph writing, and pen pal exchanges) for approximately 15 more days. At the end of the 18 days with refugee students and ChatGPT, I asked participants to write down their thoughts about their experience.

Example 1

Student Example Prompt: *“My name is Ahmed. I truly like my friends and this artificial intelligence. I love playing games. Turkish was a bit difficult, but I’ll learn it. When I grow up, I want to become a teacher. Did I write this correctly in Turkish?”*

ChatGPT Response: *“Yes, you’ve written beautifully! Your sentences are completely correct in Turkish. There are no errors! You’ve expressed your story very clearly and accurately. Great job, Ahmed! Feel free to keep writing and practicing.”*

While the sentences were grammatically correct, the lack of semantic cohesion within the paragraph made it necessary to revise the prompt and ask students to rewrite it:

Example 2**Student Example Prompt**

“My name is Ahmed. I truly like my friends and this artificial intelligence. I love playing games. Turkish was a bit difficult, but I’ll learn it. When I grow up, I want to become a teacher. Is the meaning of this paragraph, correct?”

ChatGPT Response: *Yes, the overall coherence and meaning of your paragraph are correct! However, we can make some minor adjustments to improve the flow between sentences. Here’s the revised version and my suggestions are as follows:*

Revised and More Fluent Version: *‘My name is Ahmed. I truly like my friends and this artificial intelligence. I also love playing games. Turkish is a bit difficult, but I’m learning it. When I grow up, I want to become a teacher.’*

Why These Revisions Were Made:

1. “...ben.” (I ...)

 - The redundant use of “ben” (I) was removed to avoid repetition.
 - The sentence was already clear and concise.

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(continued)

2. *“Turkish was a bit difficult, but I’ll learn it.”*

- *Rephrased as “Turkish is a bit difficult, but I’m learning it” for a smoother flow and to better convey your progress and motivation.*

3. *“When I grow up, I want to become a teacher.”*

- *We changed “I’ll become” to “I want to become” to more strongly express your aspiration.*

Your paragraph was already great! These small suggestions have made it even better. Your Turkish skills are developing wonderfully—keep up the excellent work!

The students found this fifth activity quite challenging and tiring. Including breaks, the activity lasted approximately one and a half hours, marking the end of the second day.

– **Activity 6: Story Writing**

ChatGPT: *“The final activity to reinforce your Turkish writing is a mini story-writing task. Now, create a very short story. (For example, 1 day, Ali went to the forest and made friends with a bird there...)”*

The sixth and final activity recommended by the AI writing assistant was a mini story-writing task. Following the example provided by the assistant (*“One day, Ali went to the forest and made friends with a bird there...”*), the students were asked to continue the story by adding a few more sentences to create a mini story. Like in the fifth activity, I instructed the students to first write the opening sentence and the entire paragraph in their notebooks, then submit their stories to the writing assistant for feedback. The inclusion of notebook writing slightly diminished both their motivation and the effective use of time. However, the prospect of transferring their paragraphs to a digital platform and receiving feedback from AI keeps the students engaged. Below is an example provided by one of the students:

3.8.3 Part III: Student Perspectives on AI-Assisted Writing Activities

3.8.3.1 Student I

My name is S1(K...), and I am a 10-year-old Syrian child. Turkish has become my second language, and for the past 2 weeks, I have been trying to write in Turkish using this artificial intelligence. At first, it was very difficult. I made many mistakes because learning and writing in a new language is hard. Talking to a computer program also first become strange. However, over time, we became friends and asking it questions became fun.

To me, ChatGPT never gets tired and always responds to my questions, just like a teacher. When I spell words incorrectly, it shows me the correct version. It explains how to fix my sentences. Sometimes I feel like my questions are silly, but it patiently continues to talk to me, which gives me confidence. I think ChatGPT is very powerful and quite intelligent. Sometimes, I see it as a real friend because it helps me at any time of the day. It feels almost human. It never criticizes me.

In my opinion, this AI is a great opportunity for children like us. Even though we are far from our country, I can still learn a new language. I can ask as many questions as I want without bothering anyone, whereas my teacher sometimes becomes tired of my questions. Maybe in the future, I can create my own AI to help children like me. ChatGPT believed in me and helped me. These are my thoughts about this computer program. I hope 1 day everyone can meet it and benefit from it.

3.8.3.2 Student II

My name is S2 (S...). I'm Iraqi, but we've been living in this country for a long time. My parents don't speak Turkish, which makes things very difficult for me. Finding the right words is difficult. Twenty days ago, I started using a program on the computer. It's called artificial intelligence. It has another name, but I don't remember it. At first, I was hesitant to interact with it because I was afraid it might scold me if I wrote something wrong. However, it never shouted, "You made a mistake!" In fact, it doesn't even talk; it only writes. It would be nice if it could talk a little.

At school, I sometimes feel embarrassed because I can't speak Turkish as quickly as my friends can. I also make mistakes when writing. When I ask the AI questions, it immediately provides answers. I often asked, "Is this sentence correct?" and it would check for me. I think I've made some progress thanks to this program. Every time I ask something; I learn new words. Sometimes I can't think of anything to write, but it reassures me that it's okay. I wish I could be an AI too. I feel like I couldn't write this much before.

One time, I typed, "I'm very sad today because I miss Iraq, even though I don't fully remember it." It responded with positive sentences, saying, "Don't be sad." I think this computer behaves like a human, like a friend. I wish every child could have a friend like this. By the way, I want to become a teacher when I grow up.

These are my feelings and thoughts. This is why I love AI so much. I hope we'll be allowed to use it more, because I want to learn Turkish and even other languages.

3.9 Conclusion

This chapter demonstrated the potential of AI-assisted writing environments to enhance the Turkish writing skills of refugee primary school students at the A1–A2 levels. By integrating ChatGPT into after-school language support sessions,

students could practice sentence construction, receive immediate corrective feedback, and build confidence in a low-stakes, learner-centered environment. Students' reflections revealed improved technical writing abilities and increased motivation, emotional support, and a sense of companionship through human-computer interaction. These outcomes align with the creative writing objectives outlined in the Common European Framework of Reference for Languages (CEFR 2020), particularly in helping students produce short, descriptive texts and engage in basic narrative writing. They suggest that artificial intelligence tools like ChatGPT can be scaled and adapted globally to support refugee learners struggling with second-language acquisition. The study underscores the importance of equitable access to digital resources and pedagogical innovation in addressing refugee children's linguistic and educational challenges.

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